

Visibly Observed Photocrosslinking Reaction in Indolyl Based Pendant Liquid Crystalline Polymers: Synthesis and Optical Property

KUMARASAMY GAYATHRI, CHINNUSAMY SARAVANAN, PALANINATHAN KANNAN

Department of Chemistry, Anna University, Chennai 600 025, Tamil Nadu, India

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ABSTRACT: Visibly observed photocrosslinkable pendant liquid crystalline polymers containing indolyl based chalcone were synthesized by free-radical polymerization and characterized spectroscopically. The differential scanning calorimetry and polarized optical microscopy were used to examine liquid crystalline property. The photocrosslinking and luminescence properties were monitored by UV-Vis spectrophotometer and spectrofluorimeter, respectively. It demonstrates the chalcone unit did not manifest *cis* $\leftrightarrow$ *trans*-isomerization reaction along with $2\pi+2\pi$ photodimerization upon irradiation with UV light like other chalcones hitherto reported. The photocrosslinking was visibly monitored in solution through change of fluorescent color to colorless. The spacer lengths play a key role in the reaction. The fluorescence maximum was blue shifted around 70 nm in chloroform solution upon irradiation with UV light confirms the $2\pi+2\pi$ photodimerization of chalcone unit. © 2009 Wiley Periodicals, Inc. *J Polym Sci Part A: Polym Chem* 47: 5208–5220, 2009

Keywords: conjugated polymers; fluorescence; indolyl based chalcone; irradiation; liquid crystalline polymers (LCP); optical property; photodimerization; photoreactive effects; resists

INTRODUCTION

Liquid crystalline materials have been studied for several decades in electrical and optical fields not only for scientific interests but also for their practical applications.¹ Photo-initiated polymerization has attracted long-standing interest of polymer chemists not only to understand basic mechanism and kinetics of polymerization, but also to fabricate photolithographic images for electronic and photonic applications.² Polymer film with func-

tional groups such as photoisomerizable or photocrosslinkable groups were used for optical information storage, retardation film, photo-alignment layer for liquid crystal orientation and so forth.³ Polymers containing olefinic double bond undergoes cycloaddition upon irradiation of UV light is one of the commonly used photoinduced reactions.^{4–6} Recently, it has also been noticed that photocrosslinkable liquid crystalline polymers exploited as photo-alignment layer for liquid crystals, where the photodimerization occurs by exposure to UV light and their orientation adjusted by polarization direction of irradiated light.^{7–9} Among many promising photocrosslinkable groups, chalcone has been well studied and expended in photocrosslinkable side-chain as well as main-chain polymers.^{10,11} The preparation of

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Correspondence to: P. Kannan (E-mail: pakannan@annauniv.edu)

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functional materials with chalcone moiety has attracted attention attributable to afford high sensitivity to UV irradiation.^{12–14} According to Woodward-Hoffmann rule, the photoinduced cycloaddition of two olefins to give their cyclobutane ring formation is allowed photochemically as $2\pi+2\pi$ reaction in the solid state when topological requirements such as molecular arrangements are fulfilled to minimize molecular motion referable to bimolecular reaction. Chalcone main- and side-chain polymers cross-link by $2\pi+2\pi$ cycloaddition reaction on illumination with UV light appear to possess promising photo-alignment characteristics.^{15,16} The indolyl based chalcone unit containing side-chain liquid crystalline polymers are hitherto unreported. In the present work, we report the synthesis and optical property of indolyl based chalcone containing side-chain liquid crystalline polymers. The photocrosslinking reaction of chalcone was visibly observed by changing the color in solution from fluorescent to colorless is discussed.

EXPERIMENTAL

Materials

Methanol, ethanol, phenol, THF, diethyl ether, chloroform, triethylamine, dimethylformamide (DMF) (SRL, India), and other solvents were purified by reported procedures.¹⁷ Acetone, sodium hydroxide, methacrylic acid, hydroquinone, and 2,2'-azobis(2-methylpropionitrile) (AIBN) (Merck, Germany) were used as received. 1,6-Hexanediol, 1,8-octanediol, 1,10-decanediol, 3-methylindole and 4-hydroxy acetophenone, 1,3-dicyclohexylcarbodiimide (DCC), and 4-di(methylamino)pyridine (DMAP) (Fluka, Switzerland) were used as supplied. Methacryloylchloride and *n*-bromo-1-alkanols were prepared according to the reported procedures.¹⁸ 4-(*n*-Hydroxyalkyloxy)benzoic acids (**5–7**) and 4-(*n*-methacryloyloxy alkyloxy) benzoic acids (**8–10**) were synthesized by reported procedures.¹⁹

Instrumentation

The infrared spectra were obtained on a Bruker IFS 66V Fourier transform spectrometer using KBr pellets. High-resolution ¹H and ¹³C NMR spectra were recorded on a Bruker 300 MHz spectrometer in CDCl₃ with TMS as an internal standard. The weight-average molecular weight ($\overline{M}_w$), number-average molecular weight ($\overline{M}_n$),

and polydispersity ($\overline{M}_w/\overline{M}_n$) of the polymers were obtained on a PL-GPC model 210 chromatograph using THF as eluent at 25 °C and calibrated with polystyrene standard. DSC measurements were performed at a heating rate of 5 °C/min up to 400 °C on sample taken in an aluminium pan with a pierced lid, in dry nitrogen atmosphere with an empty pan as reference. The texture of sample was observed using a Euromax polarizing optical microscope equipped with a Linkem HFS-91 heating stage and a TP-93 temperature programmer. Small quantity of sample was placed between two thin glass cover slips, heated and cooled at the rate of 5 °C/min and mesophases observed. The photographs were taken on a Nikon FM-10 camera and exposed on Kodak 200 film. The absorption spectra in spectroscopic grade chloroform solution and in thin film were measured on a Shimadzu UV-260 spectrophotometer. The photocrosslinking reaction was carried out using a 500-W high-pressure Hg lamp (Ushio SX-UI 5000) equipped with a cut filter (Sigma, UTVAF-35U) for UV irradiation (360 nm). Fluorescence spectra were measured on a Perkin–Elmer LS 45 fluorescence spectrofluorimeter with excitation near and absorption maximum of chalcone moiety, using 5 nm resolutions for both excitation and emission. The change in fluorescence of chalcone was carried out using similar filter of UV lights.

1,3-Dimethylindole (1)

3-Methylindole (9.97 g; 0.076 mol) in DMF (30 mL) was added drop wise under nitrogen atmosphere to a suspension of sodium hydride (0.076 mol, 3.04 g) in DMF (120 mL) with stirring at 0 °C for 30 min. Methyl iodide (9.5 mL; 0.152 mol) in DMF (15 mL) was introduced dropwise to the mixture at that temperature. After 2 h, temperature was brought up to ambient and stirred for 12 h. The reaction mixture was poured into ice/water mixture and extracted with diethyl ether (3 × 100 mL). The extract was washed with excess of water and dried over anhydrous sodium sulfate. The solvent was evaporated to give a pale brown oily liquid and purified by column chromatography (silica gel; ethyl acetate/hexane) to give color less oily liquid (10.13 mL; yield 92%).

FT-IR (KBr, cm⁻¹): 2917 and 3052 (–CH₃), 1616 (aromatic –C=C–), 738 (substituted aromatic).

¹H NMR (400 MHz, CDCl₃)δ: 7.71 (d, 1H, Ar), 7.23–7.42 (m, 3H, Ar), 6.94 (s, 1H, Ar), 3.84 (s, 3H, N–CH₃), 2.46 (s, 3H, –CH₃). ¹³C NMR (400

MHz, CDCl_3) δ : 137.3, 128.9, 126.8, 121.7, 119.1, 118.8, 110.4, 109.3 (aromatic), 32.8, 9.9 (aliphatic).

1-Butyl-3-methylindole (2)

Compound **2** was prepared in a similar manner as that of compound **1** using 4-bromobutane in lieu of methyl iodide.

FT-IR (KBr, cm^{-1}): 2982, 2834 ($-\text{CH}_2-$), 2917 and 3052 ($-\text{CH}_3$), 1618 (aromatic $-\text{C}=\text{C}-$), 742 (substituted aromatic). ^1H NMR (400 MHz, CDCl_3) δ : 7.69 (d, 1H, Ar), 7.25–7.43 (m, 3H, Ar), 6.89 (s, 1H, Ar), 4.46 (t, 2H, $\text{N}-\text{CH}_2-$), 2.47 (s, 3H, $-\text{CH}_3$), 0.90–1.74 (m, 7H, $\text{CH}_3-\text{CH}_2-\text{CH}_2-$). ^{13}C NMR (400 MHz, CDCl_3) δ : 137.4, 128.9, 126.7, 121.9, 119.5, 118.8, 110.4, 109.2 (aromatic), 53.8, 32.8, 21.5, 14.8, 9.9 (aliphatic).

1,3-Dimethylindole-2-carbaldehyde (3)

Compound **3** was prepared by a modified literature procedure.²⁰ Compound **1** (9.0 g, 0.062 mol) in dry DMF (10 mL) was introduced into a stirred mixture of phosphoryl chloride (6.3 mL, 70.2 mmol) in dry DMF (6.12 mL, 0.070 mol), around 25 °C–30 °C. The reaction mixture was set aside at 30 °C for 2 h on a steam bath and poured onto crushed ice. The resulting clear solution was treated with aqueous sodium hydroxide solution (11.88 g, 0.297 mol) in water (40 mL), so as to maintain an acidic condition until last portion of alkali added. The reaction mixture was boiled for 1 min and then cooled to room temperature. The product was extracted with diethyl ether, washed with water, brine and dried over anhydrous sodium sulfate. After evaporation of solvents, the crude product was purified by column chromatography (silica gel, hexane/ethyl acetate) to give yellow colored oily liquid (6.74 g; yield 84%).

FT-IR (KBr, cm^{-1}): 2917 and 3052 ($-\text{CH}_3$), 1674 ($-\text{CH}=\text{O}$), 1618 (aromatic $-\text{C}=\text{C}-$), 886 (o-substituted aromatic), 741 (substituted aromatic). ^1H NMR (400 MHz, CDCl_3) δ : 10.1 (s, 1H, $-\text{CHO}$), 7.62 (d, 1H, Ar), 7.08–7.39 (m, 3H, Ar), 3.98 (s, 3H, $\text{N}-\text{CH}_3$), 2.58 (s, 3H, $-\text{CH}_3$). ^{13}C NMR (400 MHz, CDCl_3) δ : 181.5 (aldehyde carbonyl), 139.7, 131.2, 127.3, 126.8, 121.2, 120.0, 110.1 (aromatic), 31.5, 8.5 (aliphatic).

1-Butyl-3-methylindole-2-carbaldehyde (4)

Compound **4** was prepared with 82% yield using similar procedure as described for the preparation of **3**.

FT-IR (KBr, cm^{-1}): 2981, 2832 ($-\text{CH}_2-$), 2921 and 3053 ($-\text{CH}_3$), 1676 ($-\text{CH}=\text{O}$), 1617 (aromatic $-\text{C}=\text{C}-$), 886 (o-substituted aromatic), 741 (substituted aromatic). ^1H NMR (400 MHz, CDCl_3) δ : 10.11 (s, 1H, $-\text{CHO}$), 7.66 (d, 1H, Ar), 7.08–7.39 (m, 3H, Ar), 4.48 (t, 2H, $\text{N}-\text{CH}_2-$), 2.58 (s, 3H, $-\text{CH}_3$), 0.88–1.75 (m, 7H, $-\text{CH}_2-\text{CH}_2-\text{CH}_3$). ^{13}C NMR (400 MHz, CDCl_3) δ : 181.5 (aldehyde carbonyl), 145.3, 137.2, 127.2, 126.7, 121.9, 119.5, 109.2 (aromatic), 51.8, 32.8, 20.5, 13.8, 10.2 (aliphatic).

1-(4-Hydroxyphenyl)-3-(1,3-dimethylindole-2-yl) Prop-2-ene-1-one (11)

Compound **3** (1 g, 0.0058 mol), 4-hydroxyacetophenone (0.7897 g, 0.0058 mol) were dissolved in 100 mL of ethanol. The solution was stirred at room temperature for 30 min. Then 40% NaOH solution was added dropwise to the reaction mixture for 20 min and stirred until yellow precipitate obtained. The mixture was poured into hydrochloric acid (4 N, 100 g ice) with vigorous stirring. The product was extracted with ethyl acetate, washed with water, brine and dried over anhydrous sodium sulfate. The solvent was evaporated, solid dissolved in diethyl ether and decanted to remove the oily impurities. The crude product was purified by column chromatography (silica gel, hexane/ethyl acetate) to give yellow colored solid (5 g; yield 86%).

FT-IR (KBr, cm^{-1}): 3345 ($-\text{OH}$), 2924 and 3039 ($-\text{CH}_3$), 1656 ($-\text{C}=\text{O}$), 1618 (aromatic $-\text{C}=\text{C}-$), 1587 ($-\text{CH}=\text{CH}-$), 884 (o-substituted aromatic), 741 (substituted aromatic). ^1H NMR (400 MHz, CDCl_3) δ : 7.95–8.05 (m, 3H, Ar, $-\text{CH}=\text{CH}-$), 7.62 (d, 1H, $-\text{CH}=\text{CH}-$), 7.09–7.48 (m, 4H, Ar), 6.97 (d, 2H, Ar), 3.85 (s, 3H, $\text{N}-\text{CH}_3$), 2.56 (s, 3H, $-\text{CH}_3$). ^{13}C NMR (400 MHz, CDCl_3) δ : 183.5 (carbonyl), 162.6, 139.7, 135.5, 134.1, 131.2, 127.6, 126.4, 125.3, 121.2, 120.0, 116.2, 110.1 (aromatic), 31.9, 10.8 (aliphatic).

1-(4-Hydroxyphenyl)-3-(1-butyl-3-methylindole-2-yl) Prop-2-ene-1-one (12)

Compound **12** was prepared from compound **4** following the similar procedure used for the synthesis of compound **11**.

FT-IR (KBr, cm^{-1}): 3345 ($-\text{OH}$), 2924 and 3039 ($-\text{CH}_3$), 1658 ($-\text{C}=\text{O}$), 1618 (aromatic $-\text{C}=\text{C}-$), 1587 ($-\text{CH}=\text{CH}-$), 885 (o-substituted aromatic), 741 (substituted aromatic). ^1H NMR (400 MHz, CDCl_3) δ : 7.94–8.06 (m, 3H, Ar, $-\text{CH}=\text{CH}-$), 7.63

(d, 1H, $-\text{CH}=\text{CH}-$), 7.02–7.35 (m, 4H, Ar), 6.78 (d, 2H, Ar), 3.97 (s, 2H, $\text{N}-\text{CH}_2-$), 2.76 (s, 3H, $-\text{CH}_3$), 0.85–1.75 (m, 7H, $-\text{CH}_2-\text{CH}_2-\text{CH}_3$). ^{13}C NMR (400 MHz, CDCl_3) δ : 182.1 (carbonyl), 162.6, 139.7, 130.2, 131.2, 130.74, 127.2, 126.4, 125.2, 121.3, 120.0, 118.9, 118.2, 116.2, 110.1, 109.2 (aromatic), 46.1, 20.2, 13.8, 9.56, 8.52 (aliphatic).

Synthesis of Monomer **ia**, **ib**, and **ic**

The monomer **ia**, **ib**, and **ic** were prepared from **5**. A typical procedure for the synthesis of **ia** is as follows: Compound **5** (0.9182 g, 0.0028 mol) dissolved in dichloromethane (20 mL) was added under nitrogen to a solution of **11** (0.8 g, 0.0028 mol), DMAP (1.37 g, 11.2 mmol), and DCC (0.5777 g, 0.0028 mol) in absolute dichloromethane (100 mL) with stirring. Subsequently, the reaction mixture was stirred for 24 h at room temperature. The resulting solution was filtered, washed with water, brine and dried over anhydrous sodium sulfate. The crude product was purified by column chromatography (silica gel, hexane/ethyl acetate) to give of monomer **ia** as yellow colored solid (2 g).

ia: FT-IR (KBr, cm^{-1}): 2952 and 2876 ($-\text{CH}_2-$), 2924 and 3039 ($-\text{CH}_3$), 1655 ($-\text{CH}=\text{CH}-$), 1615 ($-\text{C}=\text{CH}_2$), 885 (o-substituted aromatic), ^1H NMR (400 MHz, CDCl_3) δ : 8.07–8.16 (m, 4H, Ar), 7.64 (d, 1H, $-\text{CH}=\text{CH}-$), 6.96–7.46 (m, 9H, Ar, $-\text{CH}=\text{CH}-$), 6.10 (s, 1H, $\text{C}=\text{CH}_2$), 5.54 (s, 1H, $\text{C}=\text{CH}_2$), 4.14 (t, 2H, $\text{Ar}-\text{O}-\text{CH}_2$), 4.03 (t, 2H, $-\text{CH}_2-\text{COO}-$), 3.86 (s, 3H, $\text{N}-\text{CH}_3$), 2.58 (s, 3H, $-\text{CH}_3$), 1.39–2.16 (m, 8H, $-(\text{CH}_2)_4-$). ^{13}C NMR (400 MHz, CDCl_3) δ : 188.6, 182.5 (carbonyl), 164.3, 163.7, 154.6, 139.2, 136.5, 135.9, 132.4, 129.9, 128.2, 125.2, 124.6, 121.1, 119.9, 117.5, 109.5 (aromatic), 68.3, 64.8 ($-\text{O}-\text{CH}_2$), 33.9, 32.4, 29.2, 28.6, 25.9, 25.1, 24.9, 18.3, 11.0 (aliphatic).

Synthesis of Monomer **iaa**, **iib**, and **iic**

The monomer **iaa**, **iib**, and **iic** were prepared from **12** by employing the similar procedure as described for the synthesis of **ia** with a yield of 62%, 65%, and 63%, respectively.

iaa: FT-IR (KBr, cm^{-1}): 2951 and 2876 ($-\text{CH}_2-$), 2926 and 3038 ($-\text{CH}_3$), 1654 ($-\text{CH}=\text{CH}-$), 1616 ($-\text{C}=\text{CH}_2$), 885 (o-substituted aromatic), ^1H NMR (400 MHz, CDCl_3) δ : 8.10–8.16 (m, 4H, Ar), 7.63 (d, 1H, $-\text{CH}=\text{CH}-$), 6.96–7.46 (m, 9H, Ar, $-\text{CH}=\text{CH}-$), 6.10 (s, 1H, $\text{C}=\text{CH}_2$), 5.54 (s, 1H, $\text{C}=\text{CH}_2$), 4.14 (t, 2H, $\text{Ar}-\text{O}-\text{CH}_2$), 4.03 (t, 2H, $-\text{CH}_2-\text{COO}-$), 3.99 (t, 2H, $\text{N}-\text{CH}_2-$), 2.61 (s, 3H, $-\text{CH}_3$), 1.39–2.16 (m, 8H,

$-(\text{CH}_2)_4-$). ^{13}C NMR (400 MHz, CDCl_3) δ : 188.5, 182.5 (carbonyl), 164.6, 163.2, 155.1, 139.2, 134.9, 131.9, 130.2, 128.2, 124.6, 122.4, 120.5, 117.5, 108.8 (aromatic), 68.3, 64.8, 62.6 ($-\text{O}-\text{CH}_2$, $-\text{N}-\text{CH}_2-$), 33.9, 32.4, 30.2, 28.9, 28.1, 25.9, 25.6, 24.5, 21.6, 18.3, 11.5 (aliphatic).

Synthesis of Polymer **Ia–IIc**

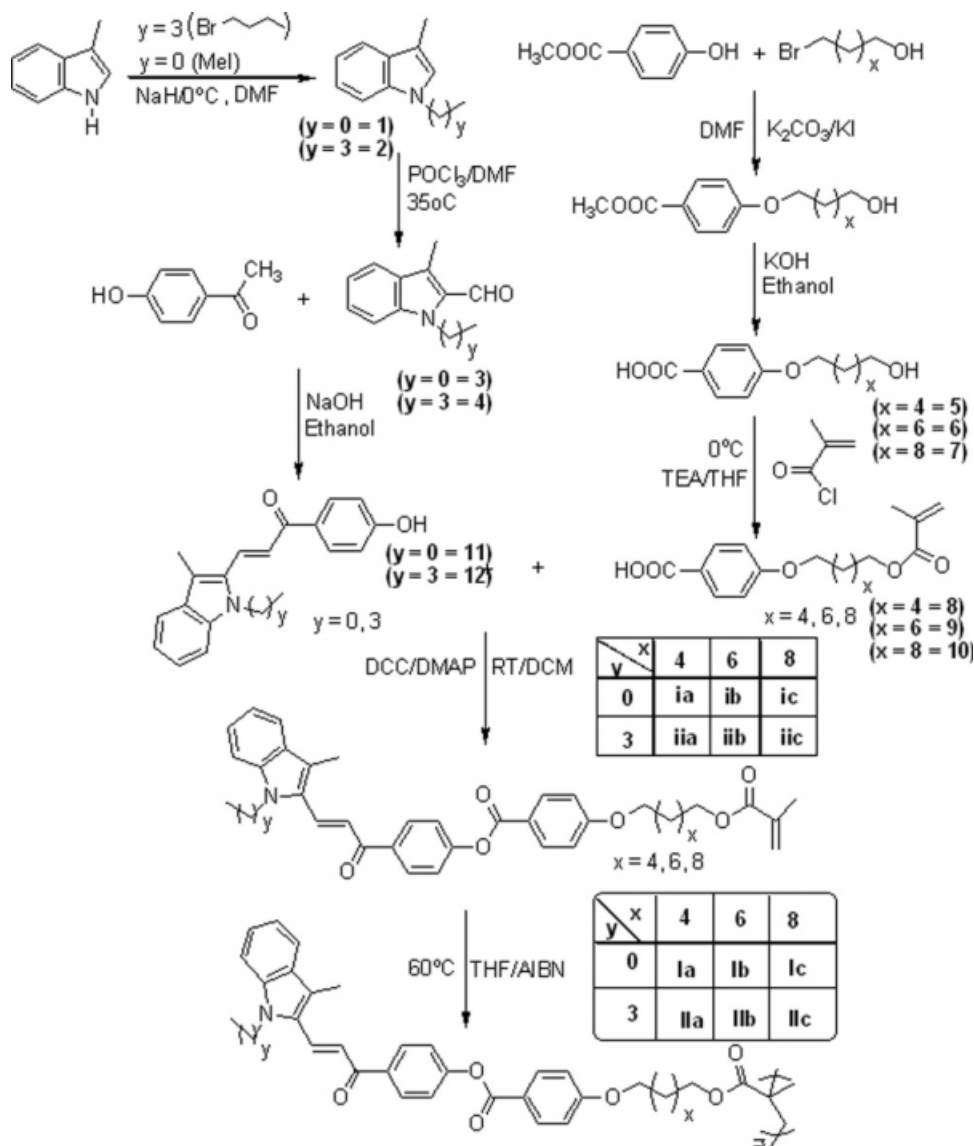
The polymers were synthesized by free-radical solution addition polymerization from the respective monomers using AIBN as an initiator in THF at 60 °C for 48 h. The typical procedure for the synthesis of polymer **ia** is as follows: The monomer **ia** (400 mg, 0.67 mol) and AIBN (2 wt %) were dissolved in dry THF (5 mL) and gentle stream of nitrogen purged into the solution. The solution was kept in an oil bath at 60 °C for 48 h. Then, the solution was cooled and poured into excess of methanol to precipitate the product. The crude polymer thus obtained was reprecipitated twice using chloroform and methanol. The purified polymer was dried at 45 °C under vacuum for 48 h to afford yellow colored solid. The remaining **Ib–IIc** was also prepared by adopting similar procedure as of **Ia**.

Ia: FT-IR (KBr, cm^{-1}): 2946 and 2881 ($-\text{CH}_2-$), 2921 and 300 ($-\text{CH}_3$), 1601 (aromatic $-\text{C}=\text{C}-$), 886 (o-substituted aromatic), ^1H NMR (400 MHz, CDCl_3) δ : 8.07 (s, 4H, Ar), 7.56–6.96 (m, 10H, Ar, $-\text{CH}=\text{CH}-$), 4.11 (s, 7H, $\text{Ar}-\text{O}-\text{CH}_2$, $-\text{CH}_2-\text{COO}-$, $\text{N}-\text{CH}_3$), 2.32 (s, 3H, $-\text{CH}_3$), 1.39–1.76 (m, 8H, $-(\text{CH}_2)_4-$). ^{13}C NMR (400 MHz, CDCl_3) δ : 183.6, 181.3 (carbonyl), 165.3, 162.7, 156.2, 135.9, 132.4, 129.9, 128.2, 124.6, 121.1, 117.5, 109.5 (aromatic), 68.3, 64.8 ($-\text{O}-\text{CH}_2$), 34.9, 33.4, 28.2, 27.6, 24.9, 24.9, 16.3, 11.0 (aliphatic).

IIa: FT-IR (KBr, cm^{-1}): 2947 and 2878 ($-\text{CH}_2-$), 2931 and 3019 ($-\text{CH}_3$), 1599 (aromatic $-\text{C}=\text{C}-$), 886 (o-substituted aromatic), ^1H NMR (400 MHz, CDCl_3) δ : 8.11 (s, 4H, Ar), 7.71–6.96 (m, 10H, $-\text{CH}=\text{CH}-$, Ar), 4.07 (s, 6H, $\text{Ar}-\text{O}-\text{CH}_2$, $-\text{CH}_2-\text{COO}-$, $-\text{N}-\text{CH}_2-$), 2.52 (s, 3H, $-\text{CH}_3$), 1.39–1.92 (m, 15H, $-(\text{CH}_2)_4-$, $-\text{CH}_2-\text{CH}_2-\text{CH}_3$). ^{13}C NMR (400 MHz, CDCl_3) δ : 183.1, 181.7 (carbonyl), 162.3, 161.7, 148.2, 143.2, 136.2, 135.9, 132.4, 129.9, 125.2, 123.6, 119.3, 118.5, 102.6 (aromatic), 68.3, 64.8 ($-\text{O}-\text{CH}_2$), 34.9, 33.4, 25.3, 24.2, 18.3, 15.2, 11.3 (aliphatic).

Preparation of the Film

Quartz slides (25 mm $\times$ 25 mm $\times$ 1.5 mm) were cleaned with water, acetone, and chloroform in an ultrasonic bath for 30 min and dried in a hot air



Scheme 1. Synthesis of polymers Ia–IIc.

oven at 60°C for 3 h. The polymer was dissolved in chloroform, filtered with 0.45 nm syringe filter, and then spin-coated at 1000 rpm on to sonicated quartz slide and annealed around their T_g to yield dry amorphous film of good optical quality.

RESULTS AND DISCUSSION

Synthesis

A series of side-chain liquid crystalline polymethacrylates containing chalcone unit in the pendant were synthesized to investigate the photocrosslinking ability of chalcone moiety. The synthetic route of monomers and polymers are presented in

Scheme 1. The chemical structure of monomers and polymers was confirmed by FT-IR, ^1H and ^{13}C NMR spectroscopy. The resonance signals for protons were assigned with alphabet as shown in the Figure (refer Supporting Information). The spectrum agrees well with targeted molecular structure of **iiia**. The ^1H NMR spectrum of **iiia** showed resonance signals at 6.88–8.14, 5.54 and 6.09, 2.21 ppm corresponding to aromatic, methacrylate vinyl, and methyl protons respectively. In addition, chalcone protons ($-\text{CH}=\text{CH}-$) resonated at 7.26–7.39 ppm as doublet. After polymerization, the resonance signal at 2.21 ppm corresponding to α -methyl protons of monomer shifted up field by 0.5 ppm and merged with resonance signal of main-chain methylene

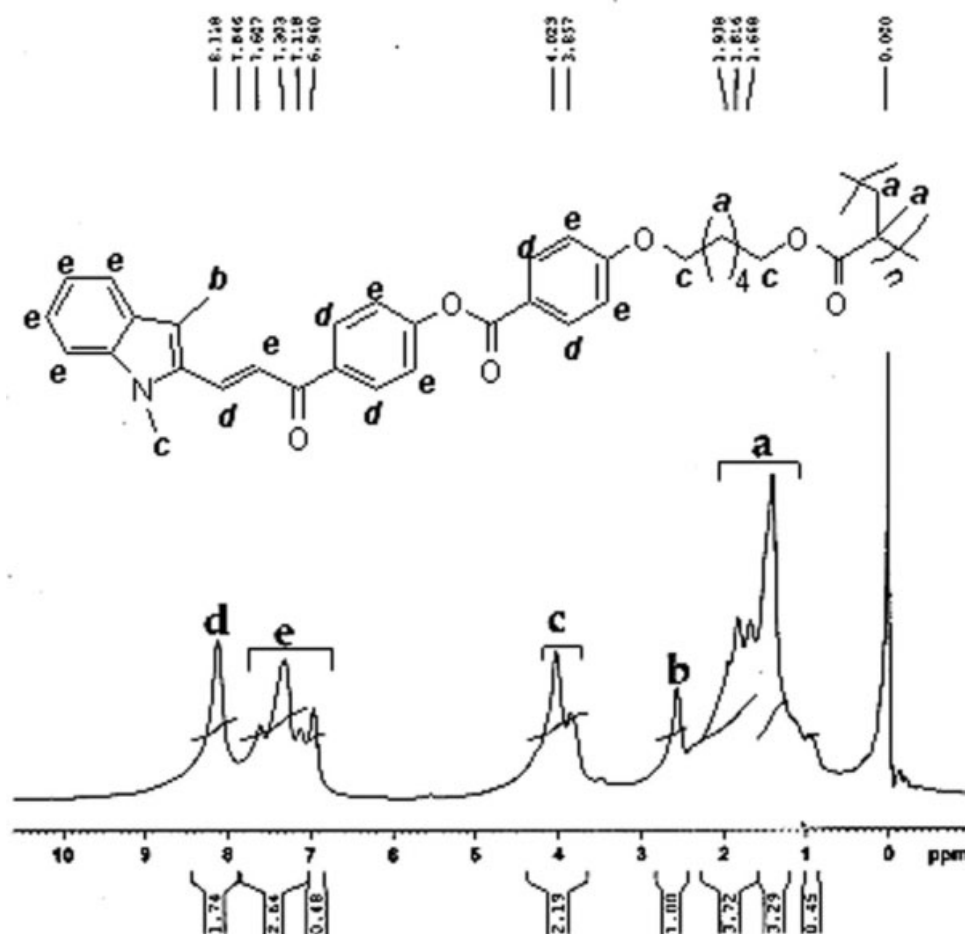


Figure 1. ^1H NMR spectrum of polymer **Ia** in CDCl_3 .

protons in the polymers appeared at ~ 1.59 ppm as broad signal. The representative ^1H NMR spectrum of **Ia** is shown in Figure 1.

In ^1H NMR spectra of polymers, the chemical shifts at 5.54 ppm and 6.09 ppm assigned to vinyl protons of monomer disappeared completely and confirm the unsaturated methacrylic function was completely involved in the polymerization reaction. The number-average molecular weight of polymers was in the range between 2.9 and 4.9×10^4 . The polydispersity index (PDI) was between

1.09 and 1.28 (Table 1). The introduction the methylene chain between backbone and mesogenic unit as well as in the terminal position solubilized the synthesized polymers readily, in common organic solvents such as chloroform, dichloromethane, ethyl acetate, and THF.

Thermal Properties

Two series of polymers were prepared as a consequence of variations in the spacer lengths (x)

Table 1. Molecular Weight, Yield, and Absorption Maximum of Polymers

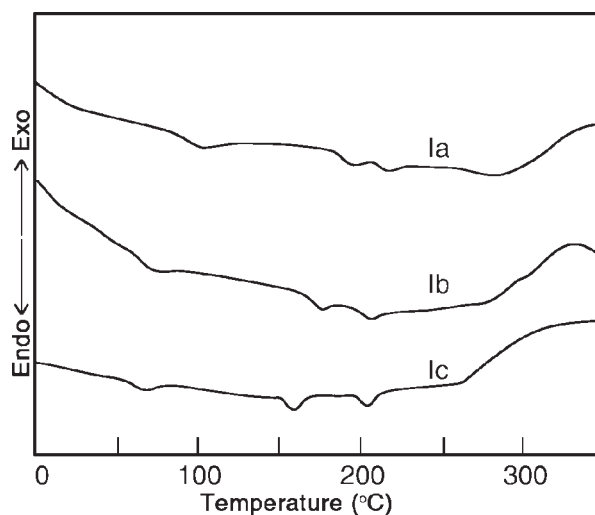
Polymer	Yield (%)	$\overline{M}_n$	$\overline{M}_n/\overline{M}_w$	λ_{max} (nm) (Chloroform)	λ_{max} (nm) (Thin Film)
Ia	67	42,600	1.21	398	398
Ib	66	41,300	1.09	397	397
Ic	71	49,200	1.18	399	399
IIa	62	32,975	1.28	406	406
IIb	65	36,254	1.17	405	405
IIc	63	29,721	1.21	405	405

Table 2. TGA Data of Polymers **Ia–Ic**

Polymer	5%	50%	% Carbine Residue at 800 °C
Ia	291	335	24.69
Ib	285	325	24.12
Ic	270	317	21.92
IIa	286	310	21.01
IIb	272	295	19.73
IIc	263	286	12.32

between the mesogen and backbone as well as the lengths of terminal alkyl chains (y). The thermal stability was estimated using TGA under nitrogen atmosphere at a heating rate of 10 °C/min. The thermograms show two stage decomposition transitions above 250 °C, indicating that all the polymers possess good thermal stability. The lower decomposition is probably resulted from cleavage of photo-crosslinking moiety of chalcone unit while higher one is suggested to be the decomposition of methacrylate backbone. The thermal stability was evaluated by 5 and 50% weight loss at minimum temperature and data are illustrated in Table 2. The methyl substituted polymers (**Ia–Ic**) shows higher thermal stability than butyl substituted polymers (**IIa–IIc**). In general, the thermal stability was decreased with increasing spacer length from 6 to 10. It may be attributed to the presence of ether linkage in the spacer units were expected to introduce greater flexibility and consequently brought down thermal stability.⁶ The decomposition was completed around 600 °C, and then no weight loss occurred. The char yield was measured at 800 °C. It shows that the char yield was increased with decreasing spacer length. In general, methyl substituted polymers (**Ia–Ic**) demonstrated higher char yield than butyl substituted polymers (**IIa–IIc**).

The liquid crystalline property of polymers was performed using DSC and POM studies. The DSC thermograms of second heating cycles of **Ia–Ic** and **IIa–IIc** are shown in Figures 2 and 3, respectively, and extracted data summarized in Table 3. A well-defined glass transition (T_g) phenomena was observed above room temperature. The DSC curve of **Ia** and **IIa** shows T_g at 98 °C and 85 °C, respectively. It suggests the introduction of methylene chain at terminal position could enhance segmental mobility to decrease T_g . Two endothermic transitions were observed on heating the sample and suggestive of transition at lower temperature corresponding to T_m and at higher temperature for liquid crystal to isotropic (T_i)

**Figure 2.** DSC thermograms of **Ia**, **Ib**, and **Ic**.

transitions. The DSC analysis revealed T_m and T_i were decreases with increasing spacers from 6 to 10. The methylene chain present on terminal position of mesogen as well plays a key role on transition temperatures. The T_m and T_i of **I** series polymers are higher than **II** series attributed to four methylene units present in the terminal position disturb arrangement of mesogen.²¹

Except **Ic**, remaining polymers exhibited uniform grainy like texture on slow cooling (0.2 °C/min) from their isotropic liquid. Whereas **Ic** shows unidentified phase at 187 °C. The POM photographs of polymer **Ib**, **Ic**, and **IIa**, **IIc** are shown in Figures 4 and 5, respectively. The phase transition temperatures observed in POM are summarized in Table 3. The alkyl chain present on

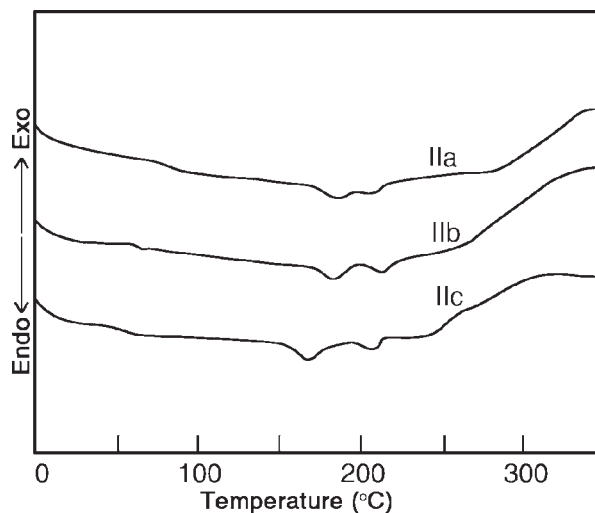
**Figure 3.** DSC thermograms of **IIa**, **IIb**, and **IIc**.

Table 3. Liquid Crystalline and Thermal Properties of Polymers

Polymer	T_g (°C)	DSC (°C)			HOPM (°C)			Mesophase
		T_m	T_i	ΔT	T_m	T_i	ΔT	
Ia	98	196	212	16	198	213	15	Grainy
Ib	75	188	208	20	189	210	21	Grainy
Ic	72	169	202	33	171	205	32	Unidentified
IIa	85	189	207	18	192	209	17	Grainy
IIb	69	179	210	31	180	211	31	Grainy
IIc	57	162	204	42	164	205	41	Grainy

terminal position alters the transition temperatures but unaffected the texture of mesophases.

Optical Properties

Photoreactive polymers containing chalcone moiety behave as a negative resist resulting from

$[2\pi+2\pi]$ photocycloaddition between an excited chalcone of one chain with ground state chalcone of another chain.^{11,22} In negative resists, the photocrosslinking occurs on irradiation with UV light renders the material less soluble in the developer; unirradiated part of coating are washed away and negative polymer image of original pattern

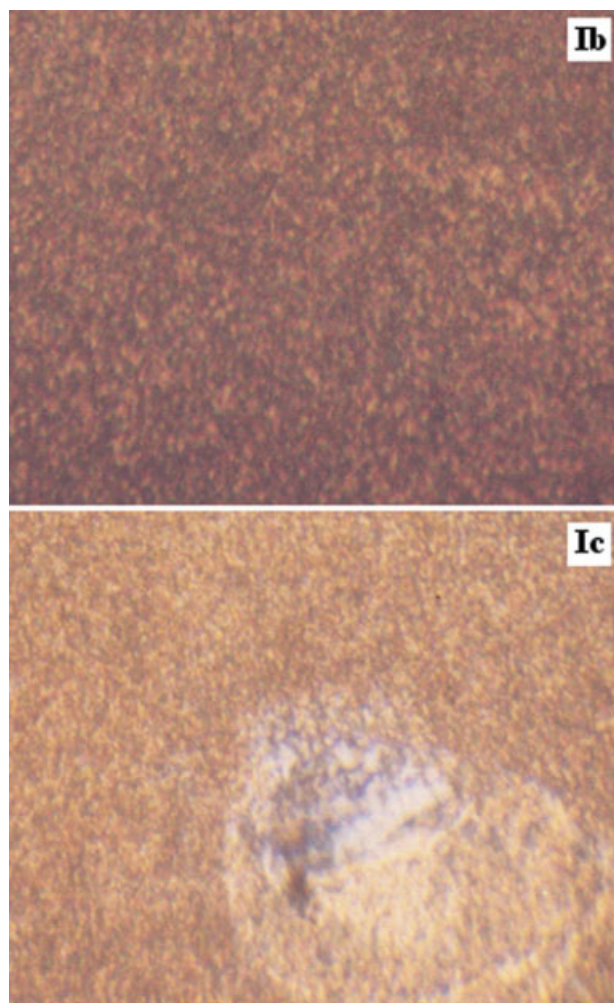


Figure 4. Representative POM photographs of **Ib** and **Ic** at 20 $\times$ magnification.

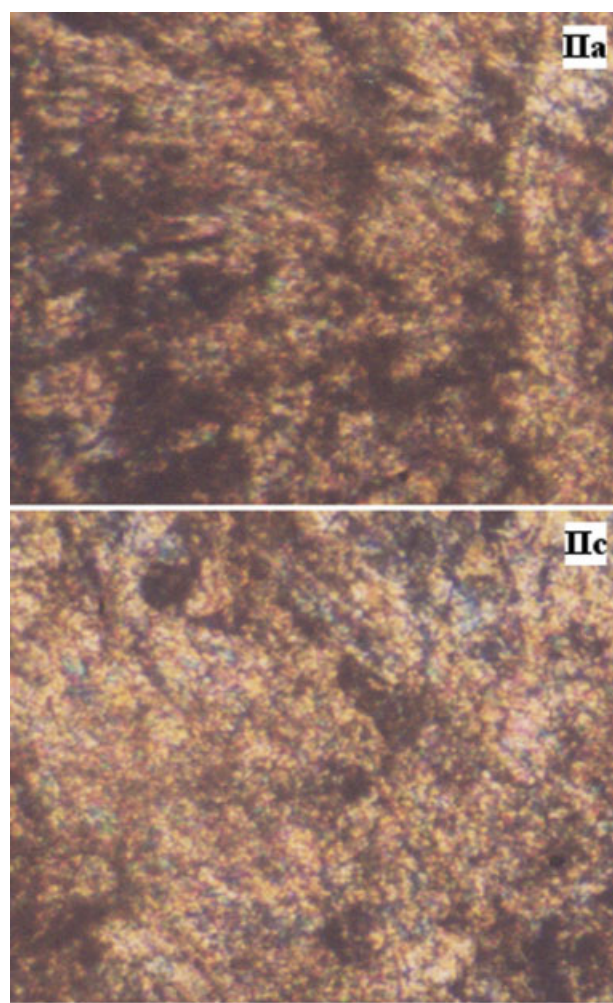
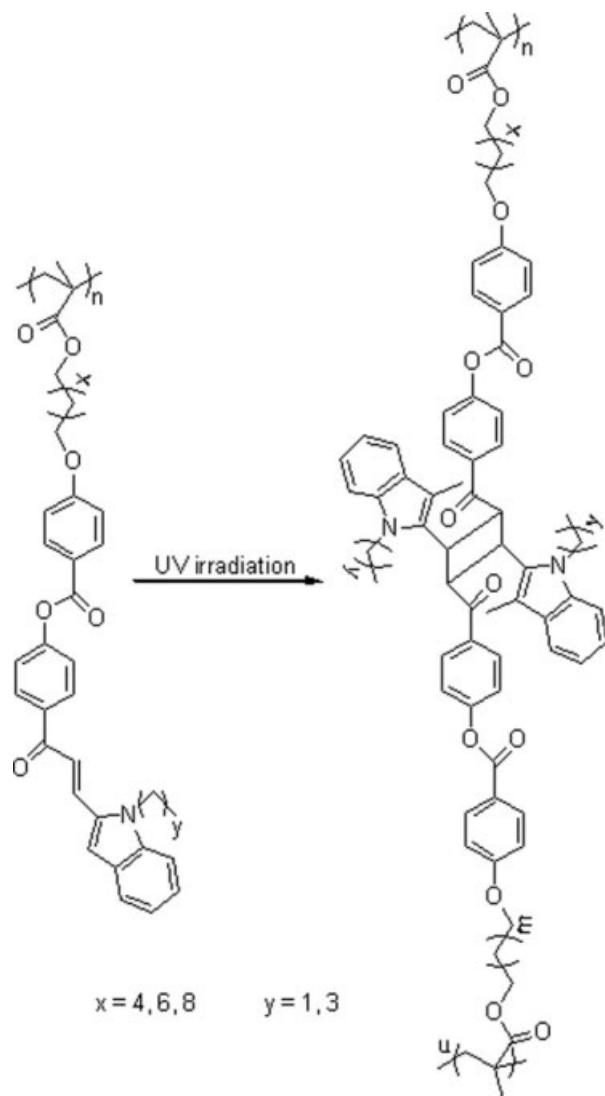


Figure 5. Representative POM photographs of **IIa** and **IIc** at 20 $\times$ magnification.



Scheme 2. Schematic representation of photocross-linking reaction of **Ia–IIc**.

remains on the substrate. The photochemical crosslinking reactions of **Ia–IIc** in solution and thin film were studied by UV irradiation at 365 nm. Figures 6 and 7 show absorption spectral pattern of **Ia**, **IIa**, and **Ic**, **IIc** in chloroform solution upon irradiation with UV light as a function of time, respectively. Before irradiation, chloroform solution **Ia–Ic** shows maxima around 398 nm and **IIa–IIc** shows around 403 nm corresponding to chalcone moiety. The absorption maxima of polymers in chloroform solution and thin film are summarized in Table 1. Upon irradiation with UV light, the intensity of absorption maximum were decreased ascribable to $2\pi+2\pi$ photocycloaddition of chalcone moiety (Scheme 2) and destroys conjugation in the entire π -electron system ensued in

decrease of intensity as shown in Figures 6 and 7. Then, the reaction system reached photostationary state resulting change of color in solution from fluorescent to colorless as shown in Figure 8. After attaining photostationary state, *in situ* turbidity appeared in solution and when it was set aside, fine particles settle down in the cuvette as an indication of crosslinking that leads to insolubility in chloroform.

In the chalcone containing polymers prepared so far, the *cis* $\leftrightarrow$ *trans*-isomerization reaction always compete with dimerization reaction of double bond upon irradiation under UV light.^{11,14,15,23} On the other hand, in the present study, absorption spectral pattern clearly indicates without a isobestic point which responsible for *cis* $\leftrightarrow$ *trans*-isomerization of chalcone double bond competing with dimerization reaction. It is attributed that the ionic interaction between two polymeric chains as shown in the Scheme 3 prevents isomerization of chalcone double bond upon irradiation with UV light. The concentration of chromophore also affects photocrosslinking reactions. When the concentration increases the time taken for decolorization increases due to ionic interactions as shown in Scheme 3. At the higher concentrations, interactions cited in Scheme 3 inhibit rate photocrosslinking reaction in chalcone double bond. At low chromophore contents, completion of decolorization taken place within 20 min.

At the same time, methyl group present in the third position may perturb close movement of chalcone unit for crosslinking reaction. The absorption spectral pattern of **Ia–Ic** and **IIa–IIc** obtained in chloroform and their relative rate of photocrosslinking are shown in Figure 9(a,b), respectively, where relative reactivity $A_0 - A_t/A_0$ is plotted against time of irradiation, in which " A_0 " the absorption before irradiation and " A_t " after irradiation at time " t ". It suggests the spacer length x and y plays a significant role in photocrosslinking reaction of chalcone moiety. When x increases from 6 to 10, the rate of crosslinking also increased and indicated in Figure 9(a). In addition to this, the butyl substituted polymers exhibited faster crosslinking rate than methyl substituted polymers and depicted in Figure 9(b). The spacer was incorporated between mesogen and backbone as well as terminal position of mesogen to increase mobility of chromophore subsequently enhances crosslinking efficiency. Further, participation of chalcone double bond in photocrosslinking reaction was proved by infrared spectroscopy. The before and after irradiated FT-

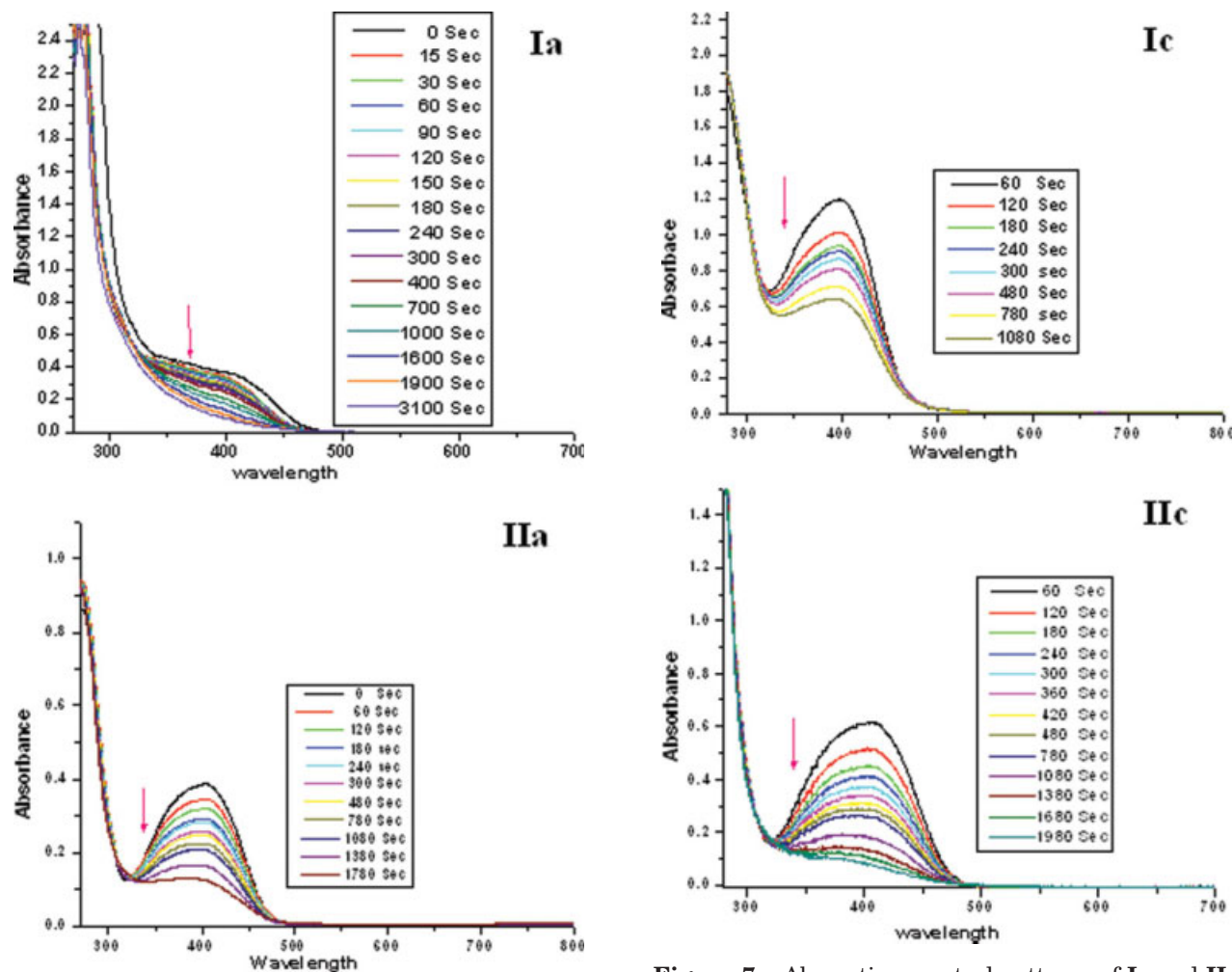


Figure 6. Absorption spectral patterns of **Ia** and **IIa** chloroform solution upon irradiation at 365 nm UV light. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

IR spectra of **Ib** is shown in Figure 10, wherein, absorption band appeared at 1653 cm^{-1} corresponding to double bond of chalcone moiety was completely disappeared after UV irradiation.

The absorption maximum and shape of the spectra for solution and film were identical for all the polymers (Table 1), ascribed to mesogenic groups did not aggregate in coated film. The representative absorption spectral pattern of polymer **Ib** and **IIb** in thin film on quartz substrates are shown in Figure 11. Upon irradiation with UV light, decrease in absorption at 397 nm and 405 nm representing to $2\pi+2\pi$ photocycloaddition of chalcone moiety in thin film of **Ib** and **IIb**, respectively. Irradiated polymer films were insoluble in chloroform as well as after irradiation color of film changed from fluorescent yellow to colorless

Figure 7. Absorption spectral patterns of **Ic** and **IIc** in chloroform solution upon irradiation at 365 nm UV light. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

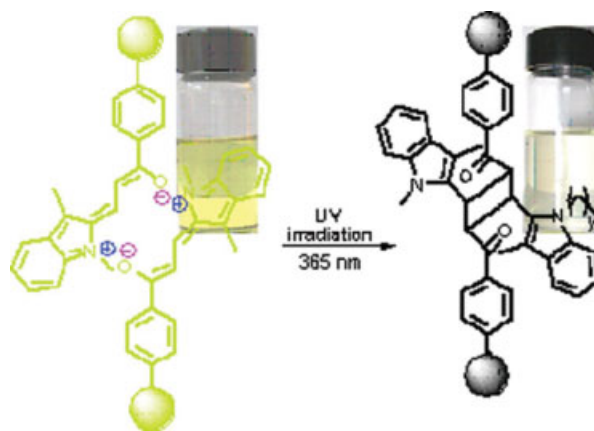
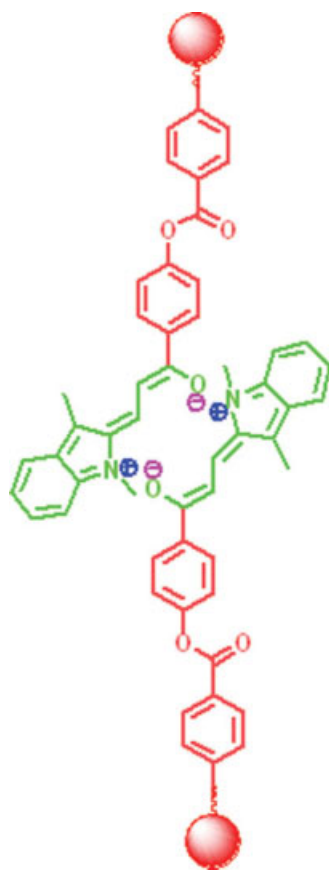


Figure 8. Representation of change in color with structure of polymer upon irradiation at 365 nm UV light. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]



Scheme 3. Schematic representation of ionic interaction between the two polymeric chains. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

confirmed the photocrosslinking reaction. In addition, the photocrosslinked polymer did not show liquid crystalline property attributed to restricted

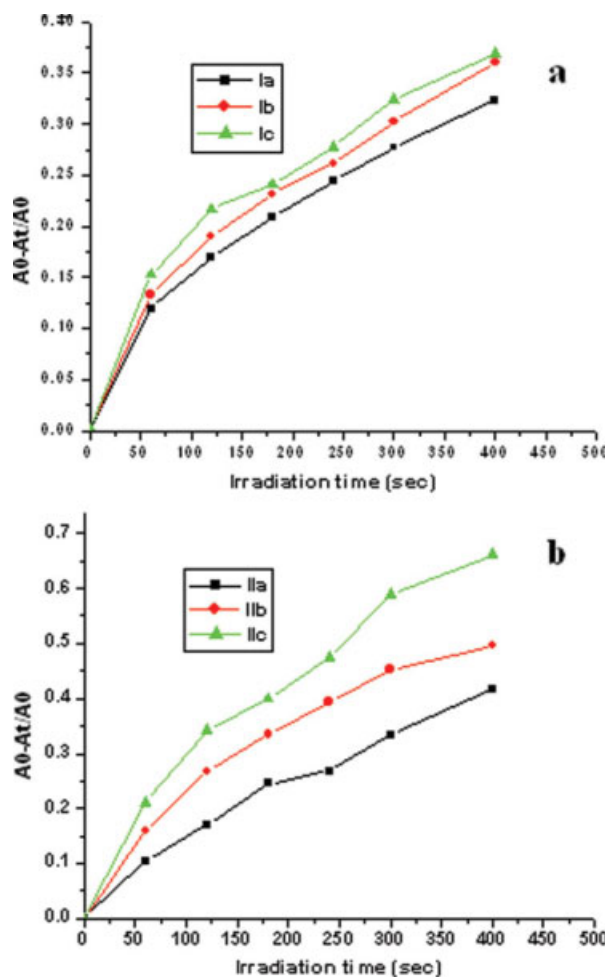


Figure 9. Dependence of photocrosslinking rate on irradiation time of polymers (a) **Ia–Ic** and (b) **IIa–IIc**. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

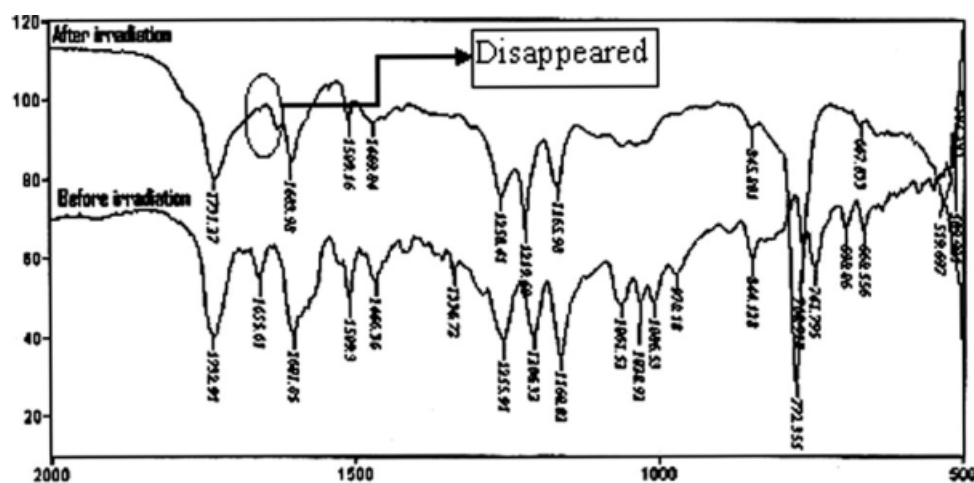


Figure 10. FT-IR spectra of **Ib** before and after irradiation of UV light.

molecular mobility and attained rigidity by the polymer backbone.

Fluorescence Property

The fluorescence emission spectra of polymers were measured in chloroform solution. They exhibited strong fluorescent emission in the range of 518–520 nm when excited with their absorption maximum due to ionic interactions as represented in Scheme 3. Upon irradiation with UV light, the absorption maximum was shifted to shorter wavelength in the range of 449–454 nm indicates the photocrosslinking reaction of chalcone moiety. The representative fluorescence spectral pattern of **Ib** upon irradiation with UV light is shown in Figure 12.

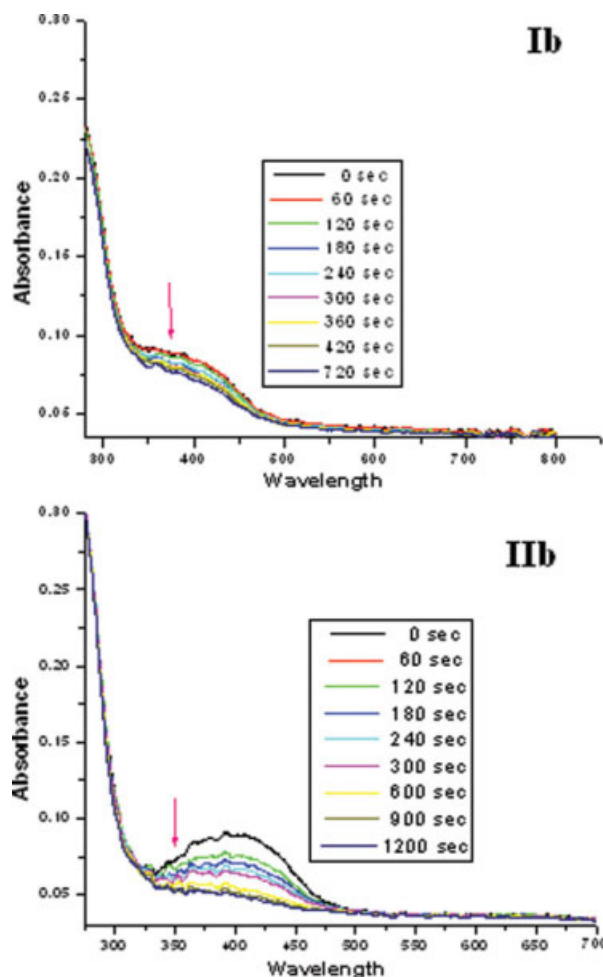


Figure 11. Absorption spectral patterns of **Ib** and **IIb** in thin film upon irradiation at 365 nm UV light. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

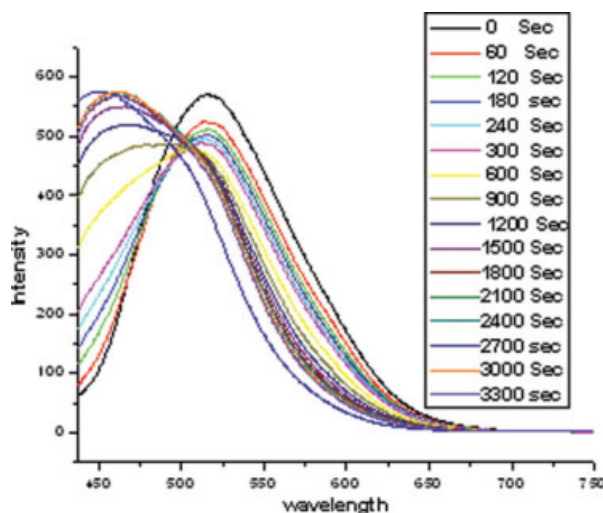


Figure 12. Change in fluorescence maxima of **Ib** in chloroform solution upon irradiation at 365 nm UV light. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

It shows fluorescence maximum at 519 nm, which was blue shifted around 70 nm and appeared at 451 nm by irradiation with UV light. After crosslinking, the fluorescence maximum appeared at 519 nm corresponding to the indole moiety. Moreover, it was noted that all the polymers showed almost similar emission maxima in the emission spectra attributed to the emission is insensitive to irrespective of spacer length between the mesogen as well as in the terminal position.

CONCLUSIONS

The polymers were prepared by free-radical solution addition polymerization and characterized spectroscopically. The POM observations show that all the polymers exhibited grainy phase except **Ic**. The polymer **Ic** exhibited nematic phase at 187 °C. The photocrosslinking reaction was monitored by UV-Vis spectrophotometer and fluorescence spectrofluorimeter. It displays the chalcone unit in the pendant is not showing *cis* $\leftrightarrow$ *trans*-isomerization reaction along with $(2\pi+2\pi)$ photodimerization upon irradiation with UV light like other chalcones reported so far. The photocrosslinking reaction of chalcone group was visibly monitored by change of fluorescent color to colorless. The spacer lengths play a key role in the photocrosslinking reaction. The fluorescence maximum was blue shifted by 70 nm in chloroform

solution upon irradiation with UV light confirms the $2\pi+2\pi$ photodimerization of chalcone unit.

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